Examiner only

2. (a) Hannah and Evan investigated the temperature increase when sodium hydroxide solution was added to dilute hydrochloric acid.

The method they used to collect their results is given below.

1. Measure 25 cm^3 of hydrochloric acid into a polystyrene cup.
2. Record the temperature of the hydrochloric acid.
3. Measure 25 cm^3 of sodium hydroxide and add it to the hydrochloric acid.
4. Record the highest temperature of the mixture.
5. Calculate the temperature increase for the reaction.

(i) Choose apparatus from the box to complete the following sentences. [2]

balance	**Bunsen burner**	**thermometer**	**measuring cylinder**
test tube	**stopwatch**	**tongs**	**beaker**

Hannah and Evan used a .... to measure 25 cm^3 of hydrochloric acid accurately.

Hannah and Evan used a .... to measure the temperature of the mixture.

(ii) Give **one** way that Hannah and Evan could check that the method produces consistent results. [1]

....

(iii) Hannah and Evan calculated a temperature increase of 17 °C.

Give the term used to describe a reaction that gives a temperature increase. [1]

....

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(b) When chlorine reacts with hydrogen, hydrogen chloride is formed.

$$H - H \quad + \quad Cl - Cl \longrightarrow \begin{matrix} H - Cl \\ H - Cl \end{matrix}$$

The bond energies are given in the table.

Bond	Bond energy (kJ)
H — H	436
Cl — Cl	243
H — Cl	432

(i) The energy needed to break the bonds in the hydrogen and chlorine molecules is 679 kJ. Show how this value is calculated. [1]

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(ii) Calculate the energy released when **two** molecules of hydrogen chloride are formed. [2]

Energy released = ... kJ

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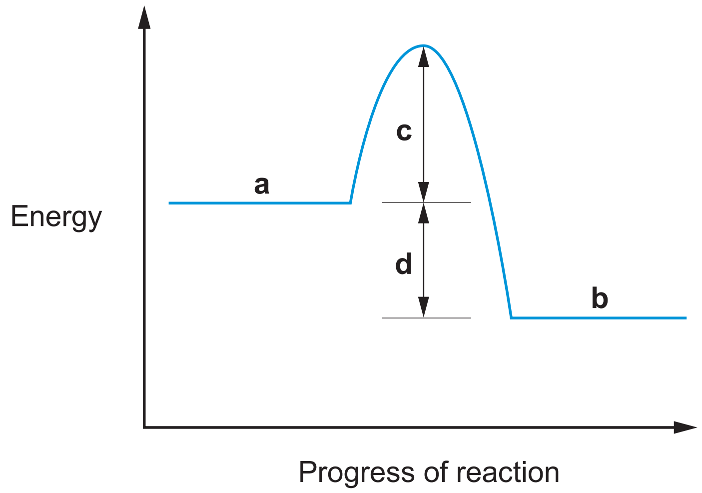
(iii) The energy profile diagram for the reaction between hydrogen and chlorine is given below.



Give the letter that shows

the activation energy ....

the overall energy change for the reaction .... [2]

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